

Naloga 20.20. Izračunajte.

- $|3 - 4i|$
- $|5 - 12i|$
- $|8 + 15i|$
- $|-20 - 2i|$
- $|0.5 + i|$
- $|\sqrt{2} + \sqrt{3}i|$
- $|\frac{3}{5} - \frac{2}{5}i|$
- $|\frac{2 - \sqrt{3}i}{2 - \sqrt{3}i}|$
- $\left(\frac{5}{|4 - 3i|}\right)^{-1}$

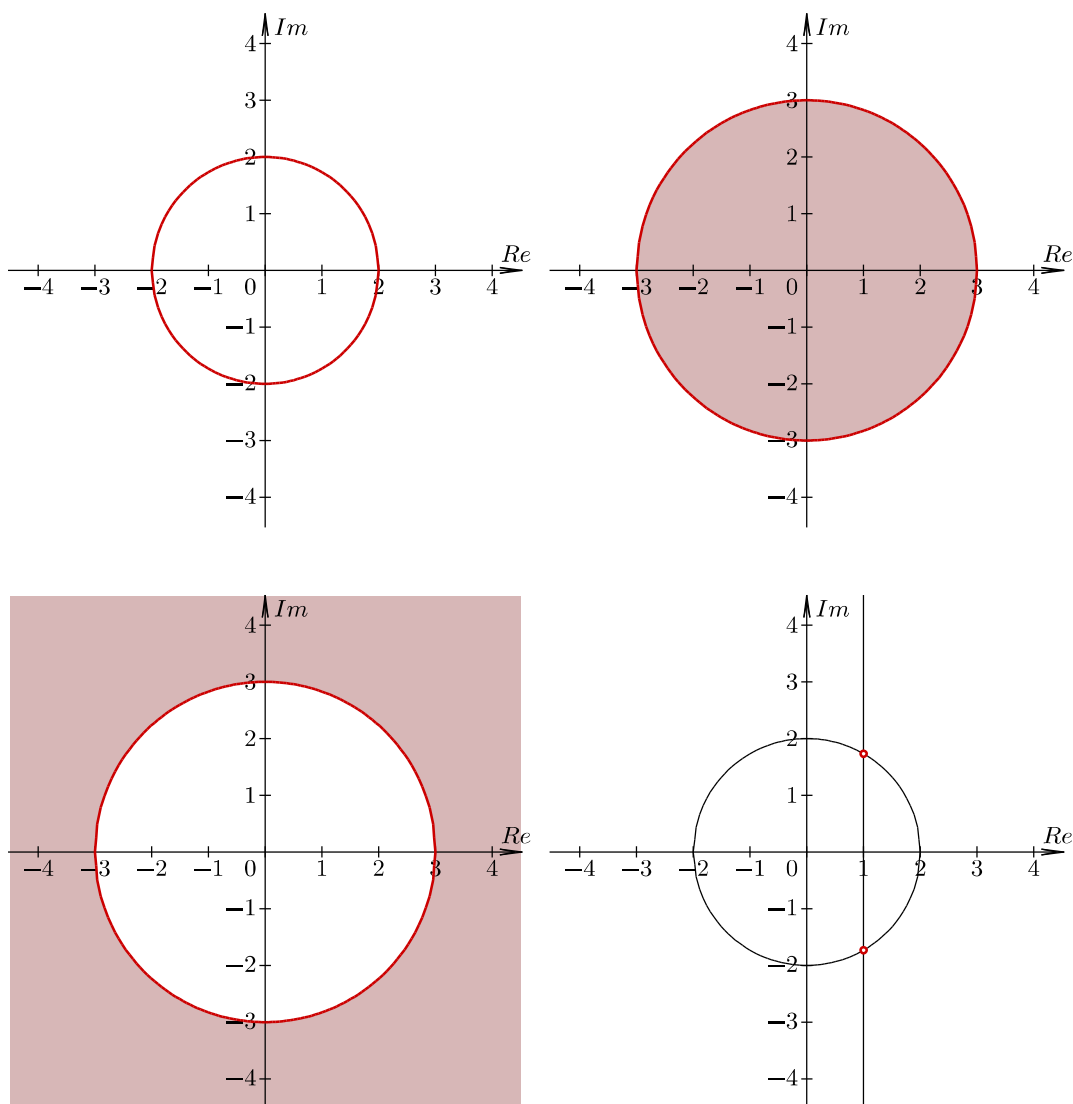
Naloga 20.21. V kompleksni ravnini upodobite množico točk, ki ustreza danemu pogoju.

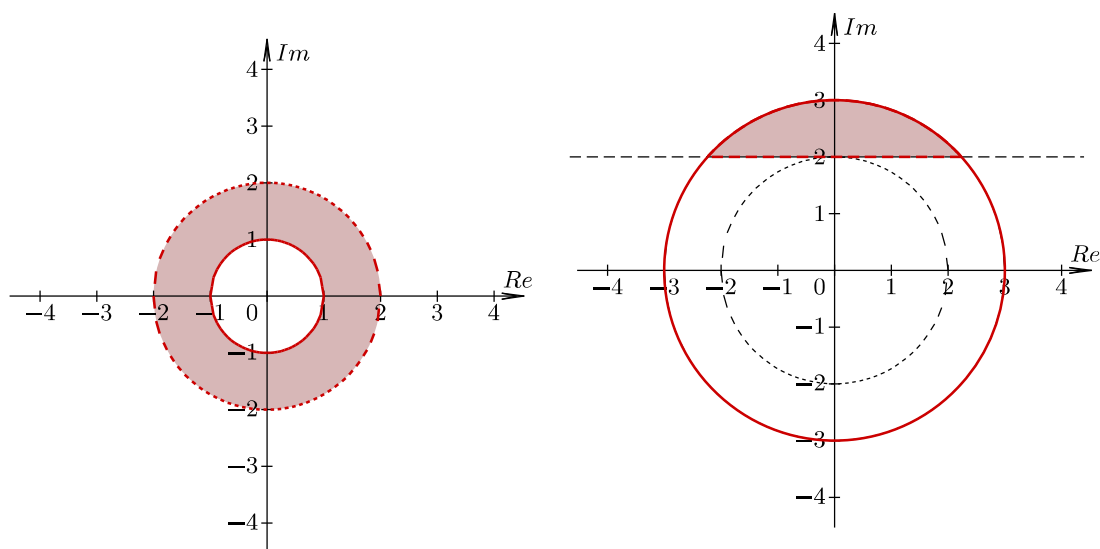
- $\{z \in \mathbb{C}; |z| < 3\}$
- $\{z \in \mathbb{C}; |z| \geq 1\}$
- $\{z \in \mathbb{C}; |z| = 2\}$
- $\{z \in \mathbb{C}; 1 < |z| \leq 2\}$
- $\{z \in \mathbb{C}; |z + 2 - i| \leq 1\}$
- $\{z \in \mathbb{C}; |z + 3 - 2i| < 2\}$
- $\{z \in \mathbb{C}; |z - 2i + 4| > 1\}$
- $\{z \in \mathbb{C}; 1 < |z - i + 2| \leq 2\}$

Naloga 20.22. V kompleksni ravnini upodobite množico točk, ki ustreza danemu pogoju.

- $\{z \in \mathbb{C}; |z| < 3 \wedge \operatorname{Re}(z) \leq 2 \wedge \operatorname{Im}(z) \geq -1\}$
- $\{z \in \mathbb{C}; |z| \geq 2 \wedge \operatorname{Re}(z) = 1 \wedge \operatorname{Im}(z) \leq 2\}$
- $\{z \in \mathbb{C}; |z| < 2 \wedge (\operatorname{Re}(z) \leq 2\operatorname{Im}(z))\}$
- $\{z \in \mathbb{C}; |z| \geq 1 \wedge (\operatorname{Re}(z) = 2\operatorname{Im}(z) - 1)\}$
- $\{z \in \mathbb{C}; |z| \leq 3 \wedge \operatorname{Re}(z) \leq 2\}$
- $\{z \in \mathbb{C}; |z| > 3 \vee (\operatorname{Im}(z) > \operatorname{Re}(z) - 2)\}$
- $\{z \in \mathbb{C}; |z| = 2 \wedge \operatorname{Re}(z) = 2\}$

Naloga 20.23. Za označeno množico točk zapišite, kateremu pogoju ustreza.





Naloga 20.24. *Izračunajte.*

- $\frac{3+i}{3+i} - \frac{26(3+i)}{2-3i} + \frac{\sqrt{-40}}{|3+i|} - i^{62}$
- $10 \cdot \frac{3-i}{2-i} + i^{2026} + |1-2i| \cdot \sqrt{-45}$
- $\frac{10}{2+i} + (1-i)^3 - i^{46} + \sqrt{-36} + |3+4i|$
- $\left| (2+i)^2 \right| - (1+3i) \overline{1+3i} + \frac{25i}{3-4i} + 3i^{44}$
- $(2+7i) \overline{(1+i)^{-1}} - \frac{5}{4+3i} - \frac{i^{93}}{5} + \left| \frac{\sqrt{3} + \sqrt{2}i}{5} \right| - \frac{7+2i}{5}$
- $10 \cdot \frac{3-i}{4+3i} + i^{203} + |-1+4i| \cdot \sqrt{68}$
- $\frac{25+50i}{3-4i} + 2i^{1624} - 3i^{15} + |\sqrt{3}-i|$
- $\left| \frac{-2-3i}{2+i} - 6i^{723} + 34(5+3i)^{-1} \right|$
- $\frac{4-5i}{4-5i} + \frac{3-2i}{3-2i} + \sqrt{52} - \sqrt{-9} + \sqrt{20}$